

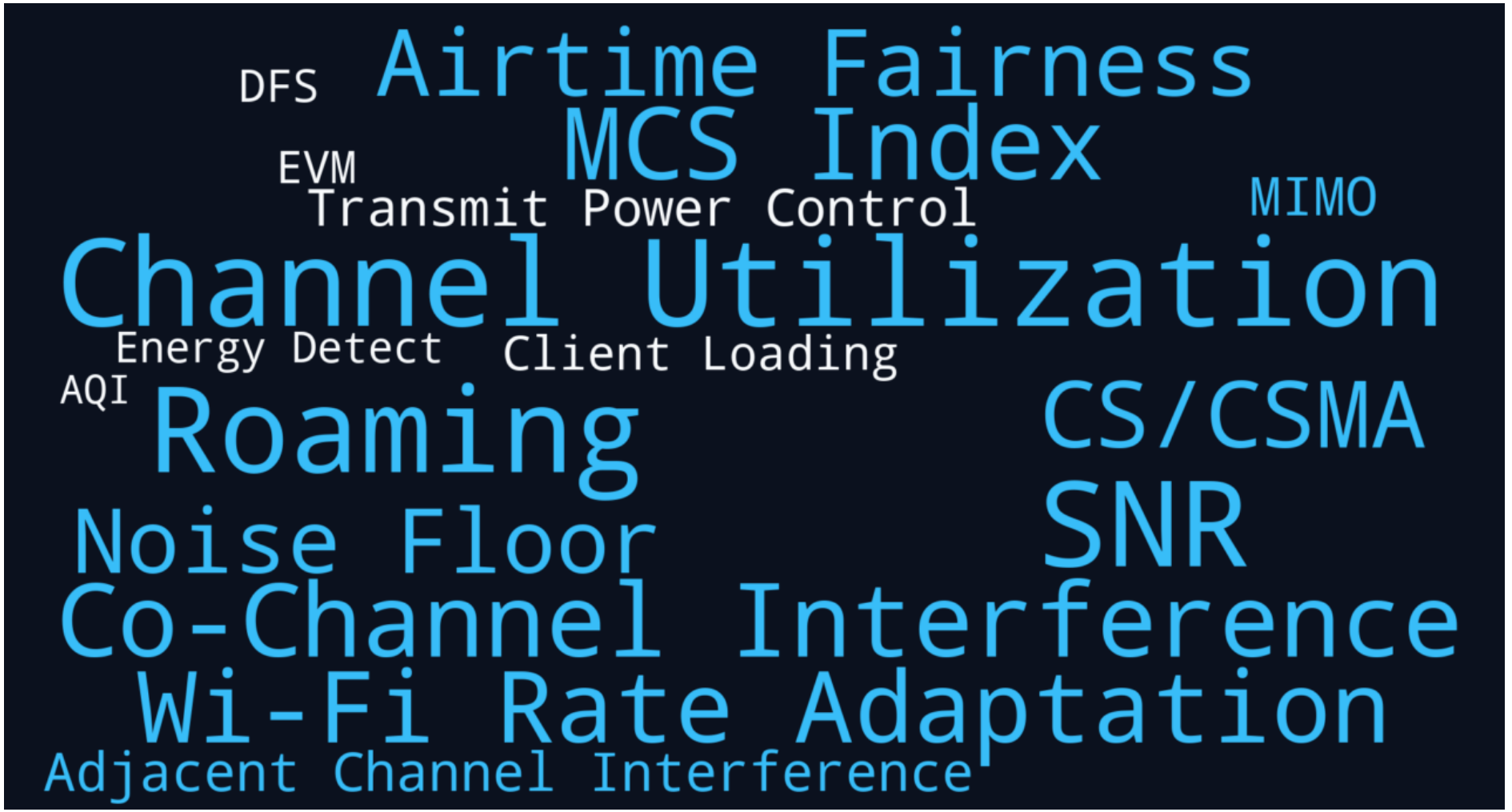
From Transistors to Throughput

How Silicon design dictates Wi-Fi performance

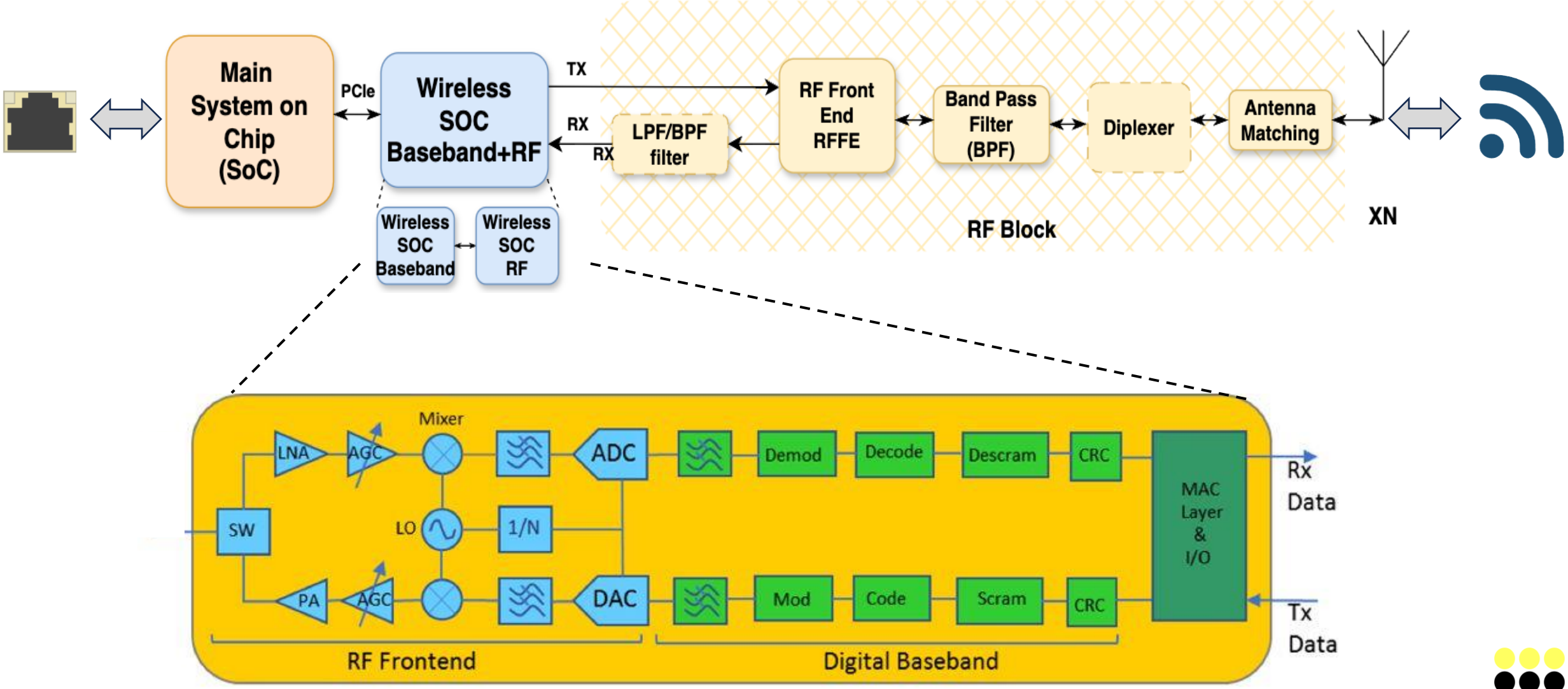
Wi-Co Montreal, Canada

7th October 2026

www.sarodeoverwireless.com

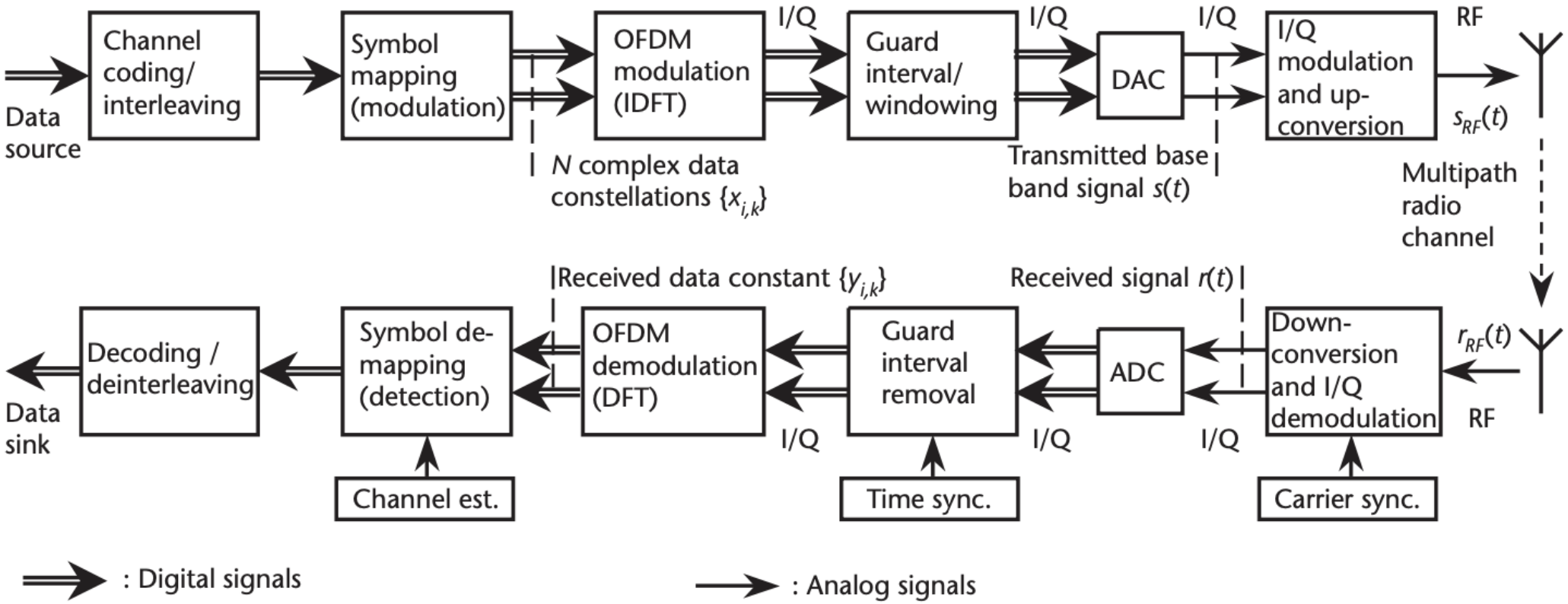

DFS
Airtime Fairness
MCS Index
EVM
Transmit Power Control
MIMO
Channel Utilization
Energy Detect
Client Loading
AQI
Roaming
CS/CSMA
SNR
Noise Floor
Co-Channel Interference
Wi-Fi Rate Adaptation
Adjacent Channel Interference

Wi-Fi signal flow



Reference: https://www.mathworks.com/discovery/wireless-transceiver.html

Digital baseband processing



Reference: IEEE 802.11 standard

RF design: Foundations of WLAN silicon



Wi-Fi 7 (802.11be) SoC

RFIC / ANALOG FRONT-END

- 4x4 MIMO RF Channels (2.4/5/6 GHz) [T/R SW, LNA, PA, PAD, Mixers]
- Dual Synthesizers (0/1) [LOGEN & VCO]
- XO & F-Doubler
- Analog LPF Bank (TX/RX Shaping)
- High-Speed ADC/DAC (320MHz Sampling)
- RF Calibration & Compensation (IQ, DC, LO Leakage)

PHY / DIGITAL BASEBAND

- Multi RU Assignment

TX PATH
- Scrambler & Puncturer
- LDPC Encoder
- Bit Interleaver
- 4096-QAM Constellation Mapper
- Spatial Mapper & Precoder
- Tone Mapping/OFDMA RU mapping & Pilot Insertion
- IFFT + Cyclic Prefix
- Digital Pre-Distortion (DPD)

RX PATH
- AGC & Timing Synchronization
- Preamble Detection & Decoding
- FFT engine (upto 4096-Pt)
- Channel Estimation & Tracking
- Frequency Domain Equalization
- Phase Noise Tracking
- 4x4 MU-MIMO Detector
- QAM Demapper Deinterleaver
- LDPC Decoder
- Descrambler & Depuncturer
- CSI Processor & Beamforming FB
- Rate Adaptation Engine

MAC SUBSYSTEM

- MULTI-LINK OPERATION (MLO) Link Steering & Aggregation
- Unified Upper MAC (U-MAC)
- A-MPDU / A-MSDU Aggregation
- Block ACK Management (512)
- Lower MAC Link 3 (L-MAC)
- EDCA & Trigger-Based Scheduling
- QoS Scheduler (Multi-AC)
- Frame Encryption/Decryption (AES-GCMP-256)
- Power Save & TIM Management
- Restricted TWT Scheduler
- Multi-RU Assignment Logic
- Preamble Puncturing Engine
- MAC Controller / Sequencer
- Descriptor Ring Management
- TX/RX Queues

SYSTEM & HOST INTERFACE

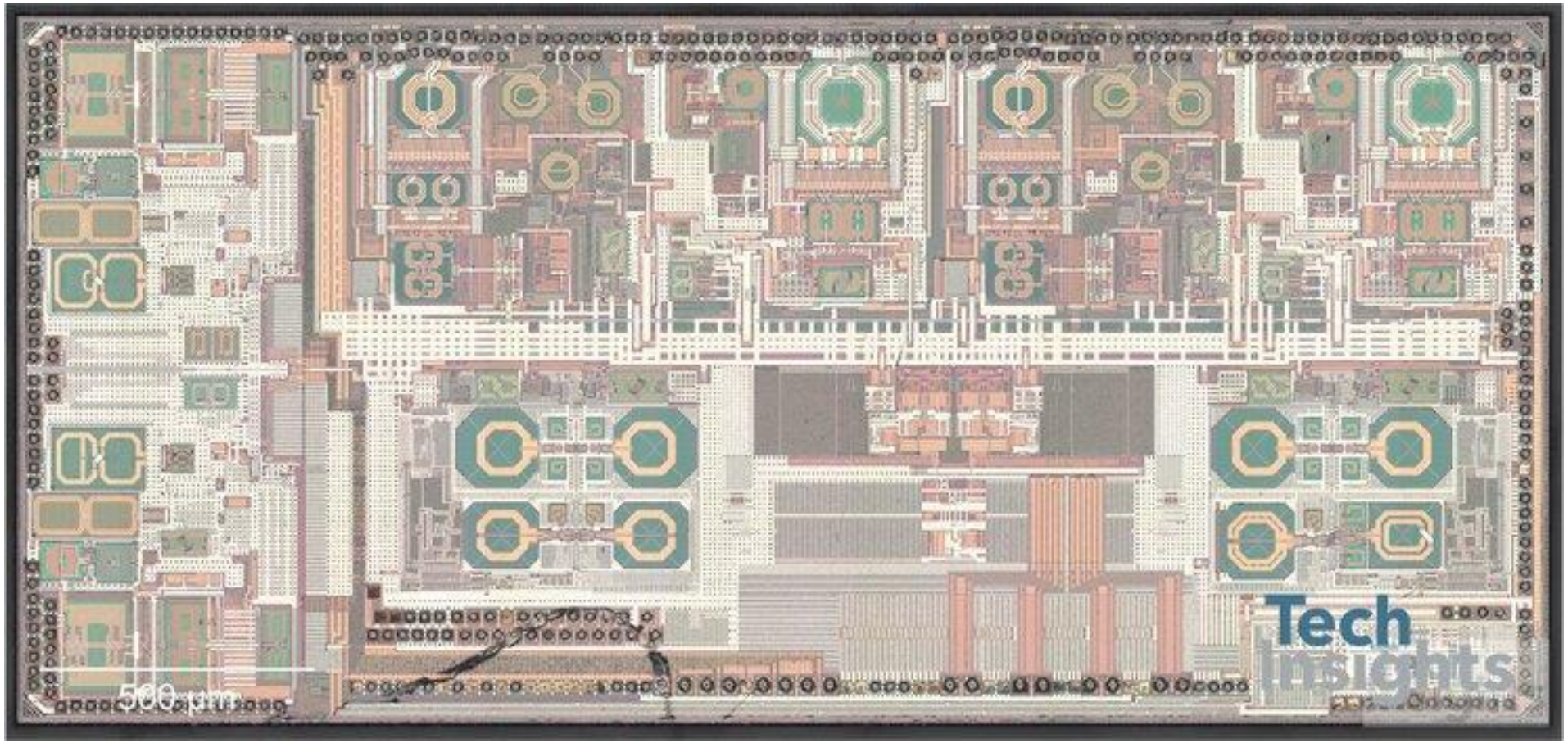
- Control CPU ARM based
- TX DMA Engine(s)
- RX DMA Engine(s)

PRIMARY HOST INTERFACE
- PCIe Gen3/Gen4 (x1/x2)

--- OR ---
- USB 3.0 (Alternate)
- Debug: JTAG, ETM

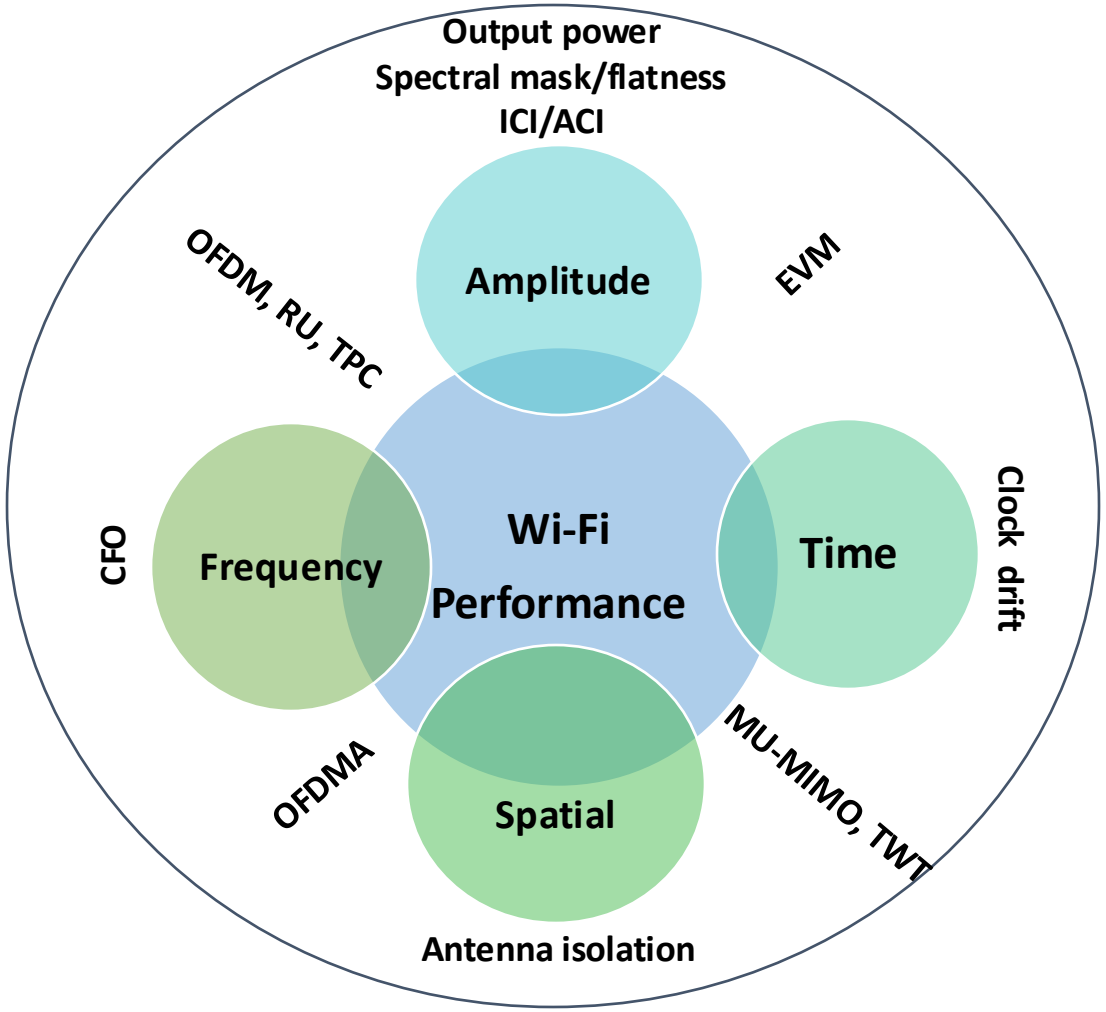
MEMORY OPTIONS
- Option A: Large Internal SRAM (4-8MB) + Host Memory via PCIe
- Option B: DDR4/DDR3L Controller + External DRAM (256-512MB)
- Flash Controller (SPI NOR/NAND for Firmware)
- Security Engine Secure Boot, Crypto Accelerator
- GPIO / UART / I2C / SPI
- Clock Generation & Distribution PLL / Clock Gating
- Power Management Unit Fine-Grained Domain Control

Wi-Fi RFIC layout



Reference: https://www.techinsights.com/blog/who-made-our-top-picks-iot-connectivity-2022

Wi-Fi operating domains



Building blocks of Wi-Fi

SNR

Signal-to-Noise

RSSI, Rate adaptation, MCS index

Noise

Ambient Thermal / CCI

Clear Channel Assessment baseline

CFO

Carrier Frequency

Uplink OFDMA subcarrier tracking.

EVM

Error Vector Magnitude

4096-QAM constellation integrity.

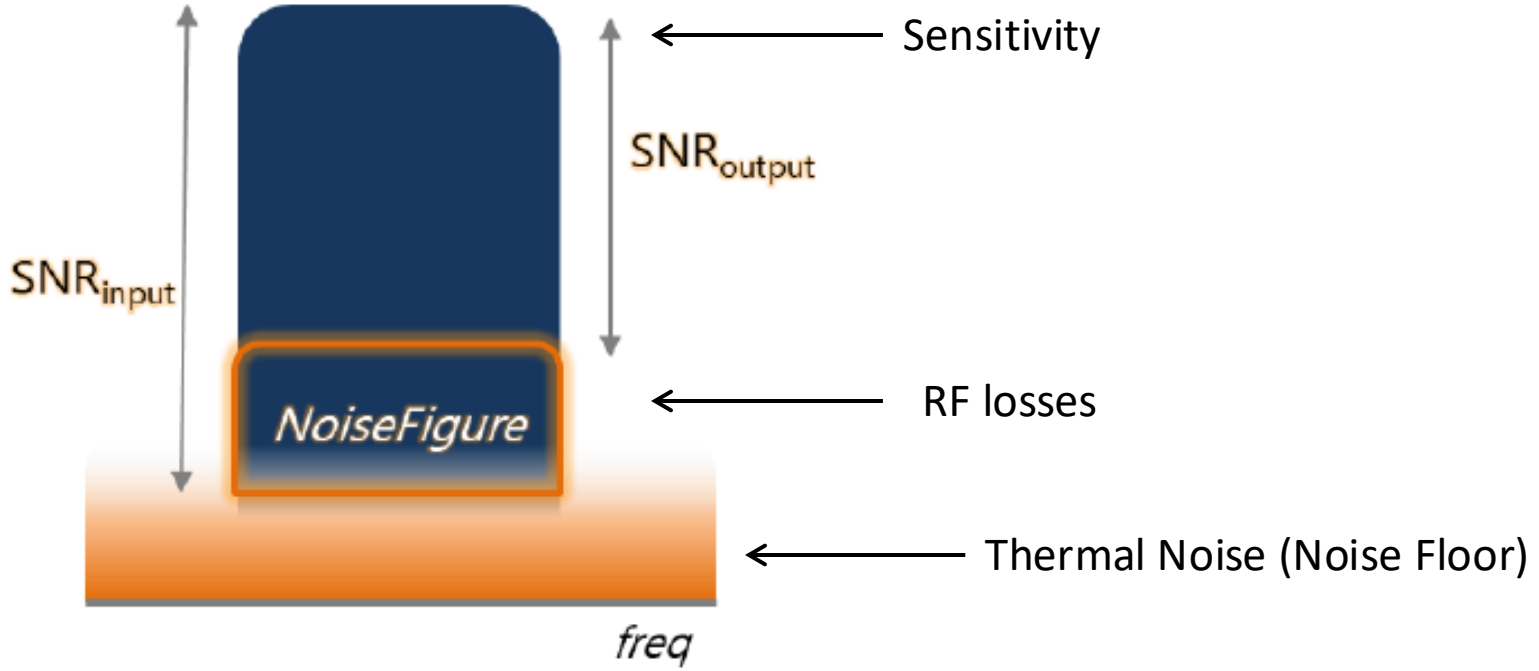
Transmit Power

RF Output Level

Coverage & Interference



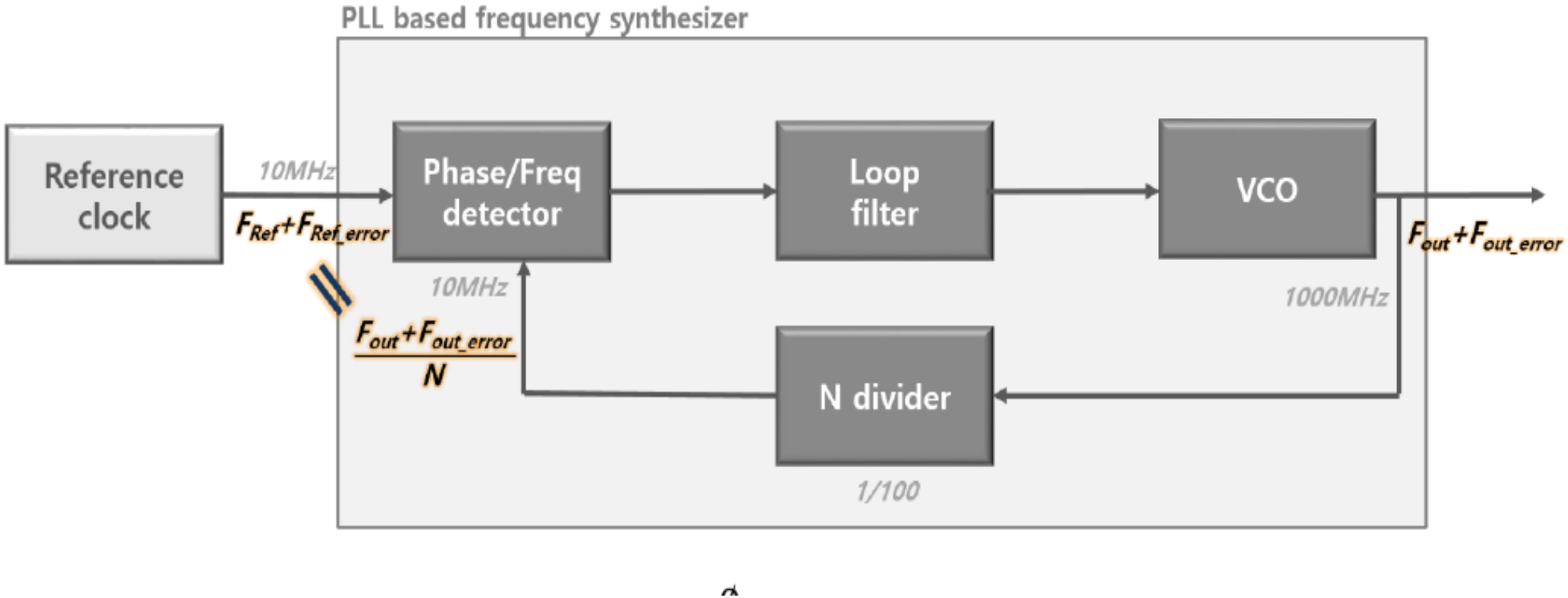
Understanding SNR & Noise



Noise : Inherent to Wi-Fi silicon design and operating bandwidth

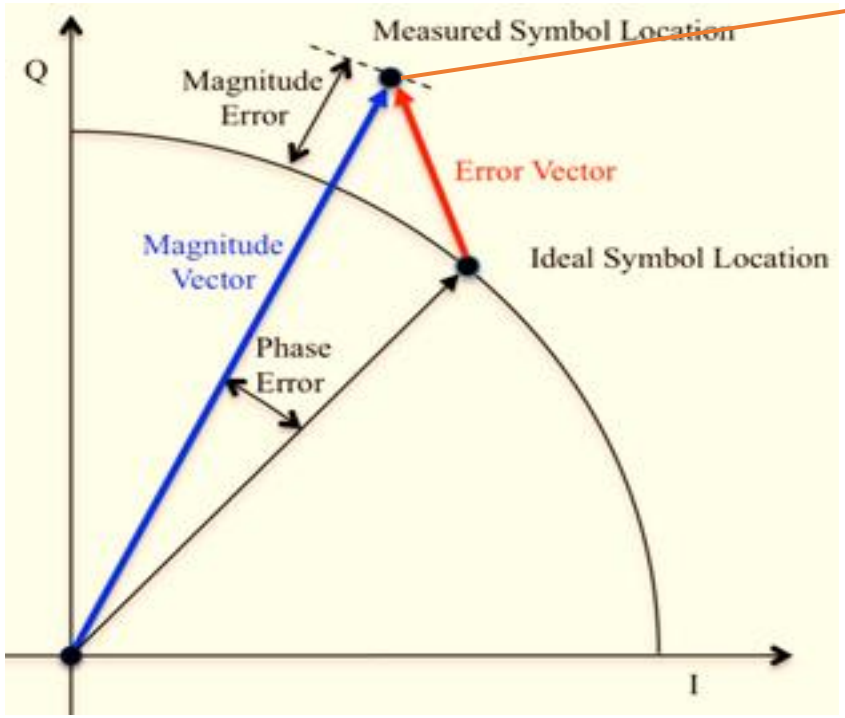
Interference : Environment dependent. Depends on Wi-Fi silicon's capability to reject interference

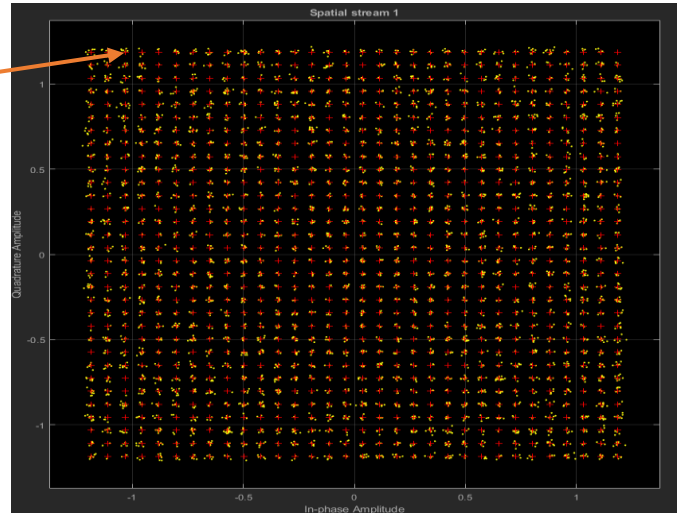
Understanding CFO



Reference clock drifts due quality of crystal oscillator, thermal design, ageing and doppler shift

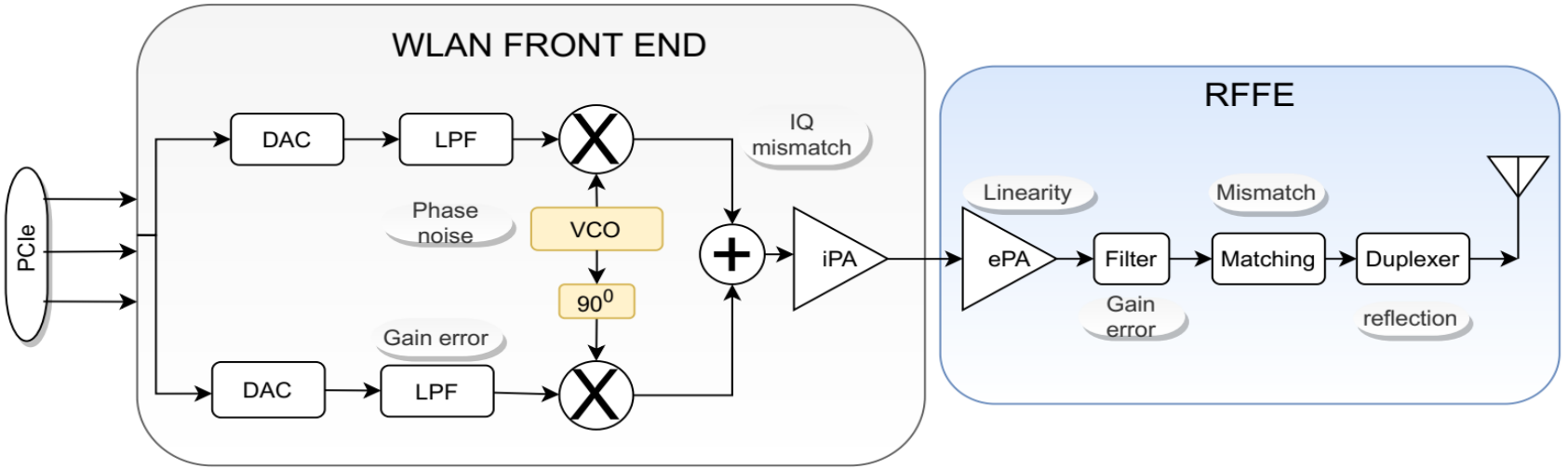
Understanding EVM

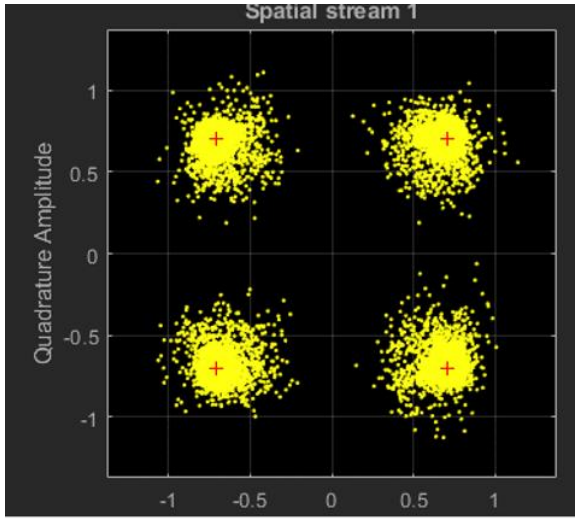


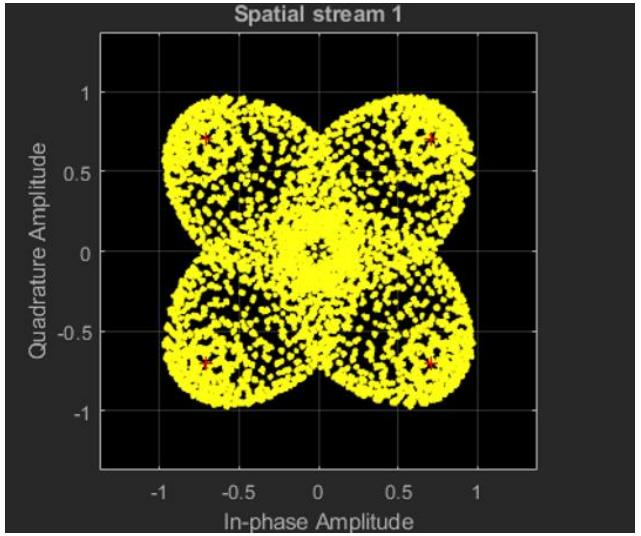


EVM	-38 dB (MCS12)
CFO	2.4 GHz: +/- 25 ppm 5/6 GHz: +/- 20 ppm
SNR	35 dB

What causes EVM

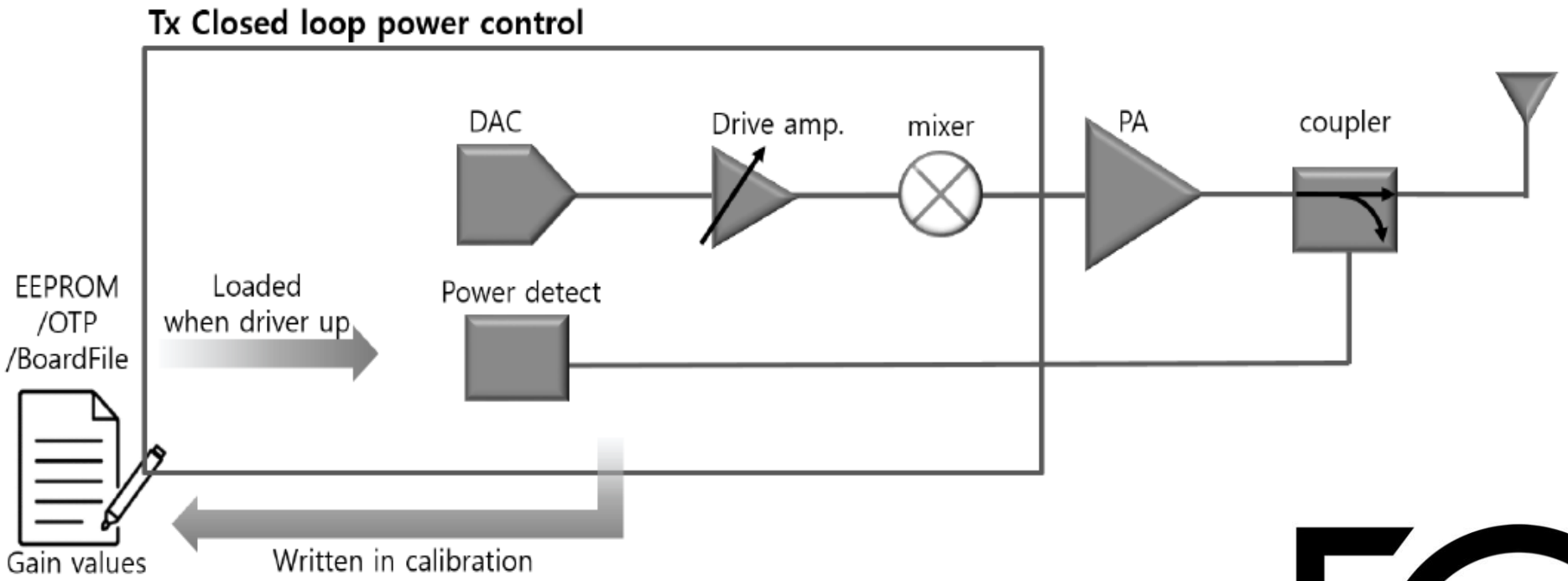






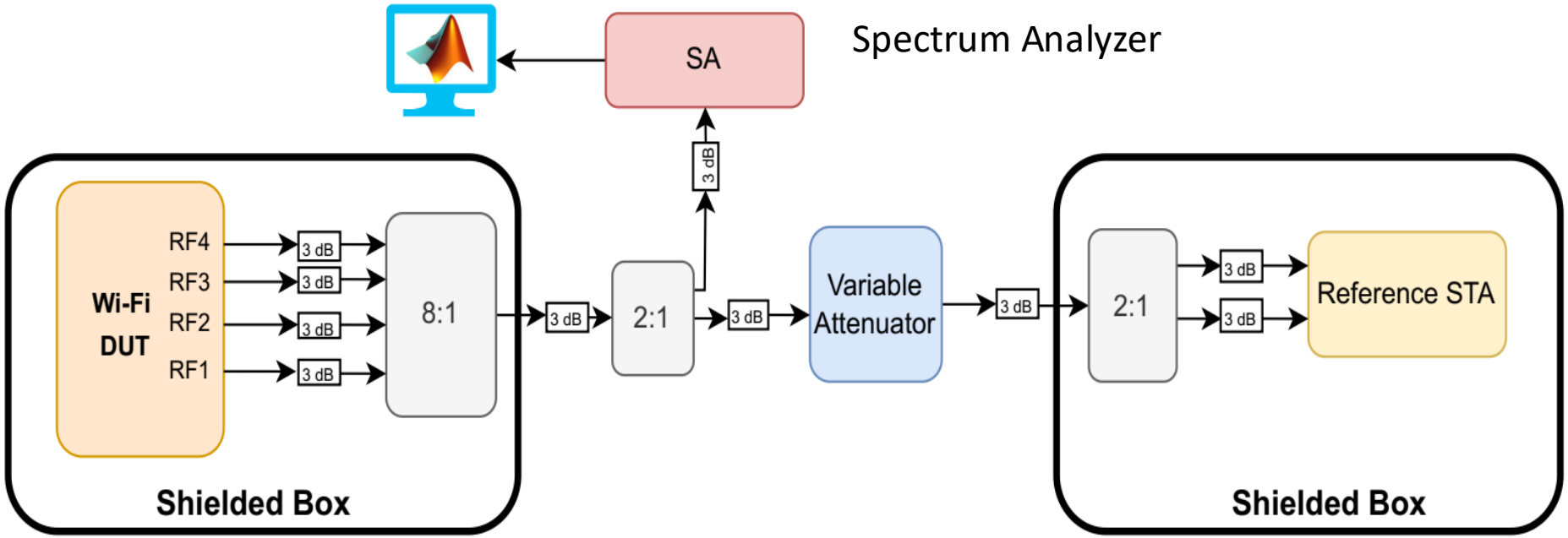
Graphic source: wlanpedia

Transmit Power



How to measure

Poor man's test setup



- Built with off-the-shelf RF components, AP, STA and no special firmware
- EVM measured on real TCP packets sent via Iperf

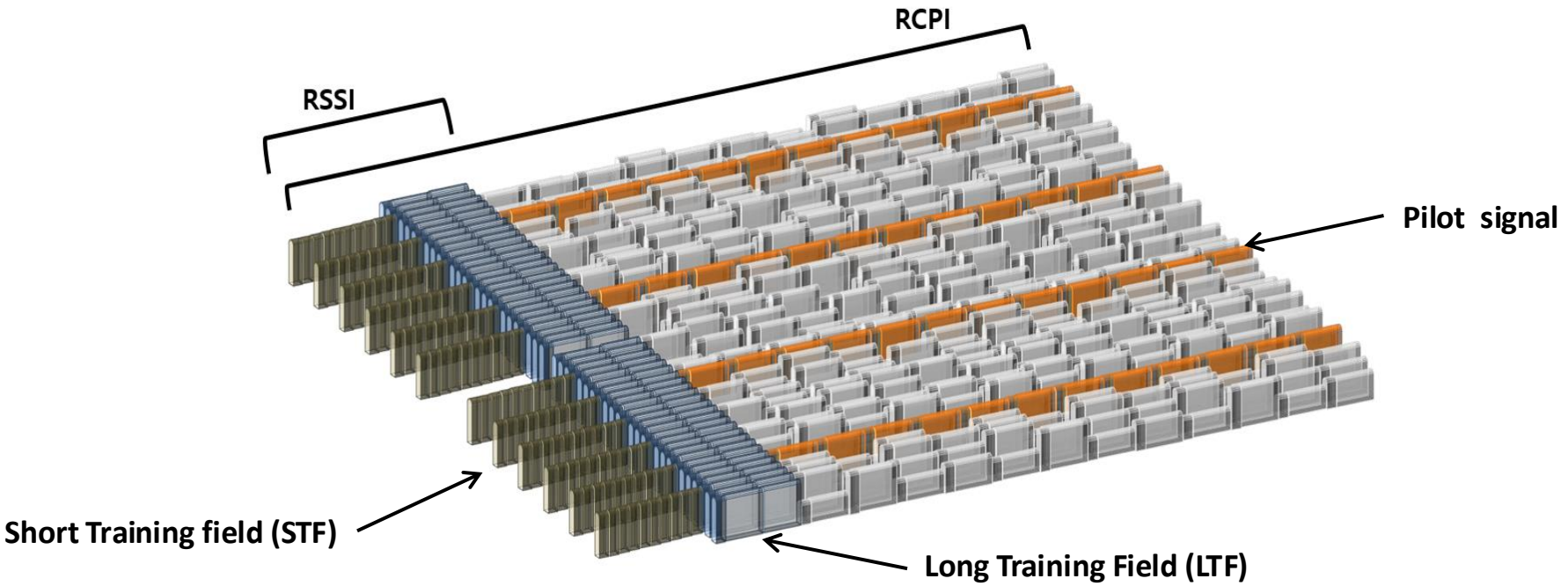
https://www.mathworks.com/products/wlan.html

The Wi-Fi workhorse

RSSI	Channel Utilization	OFDM & OFDMA	MIMO & MU-MIMO	MLO



Wi-Fi RSSI & RCPI & ANPI



Received Signal Strength Indicator (RSSI)

1 byte register, (usually) measured on STF/LTF

Received Channel Power Indicator (RCPI)

Received Power of a frame averaged over multiple RX chains

RSSI = SNR - ANPI

Average Noise Power Indicator (ANPI)

Part of 802.11K RRM, measure of channel Noise + Interference

Key RSSI Calculation Parameters

Antenna Chains

Individual or combined

Channel Width

20 MHz primary or scalable

Frame Selection

Beacon, Control or Data frames

Hardware Offset

Non-volatile board loss correction

Calculation method

Time averaged or smoothed

Accurate RSSI signal values require combined per-chain calibration.



Channel Utilization

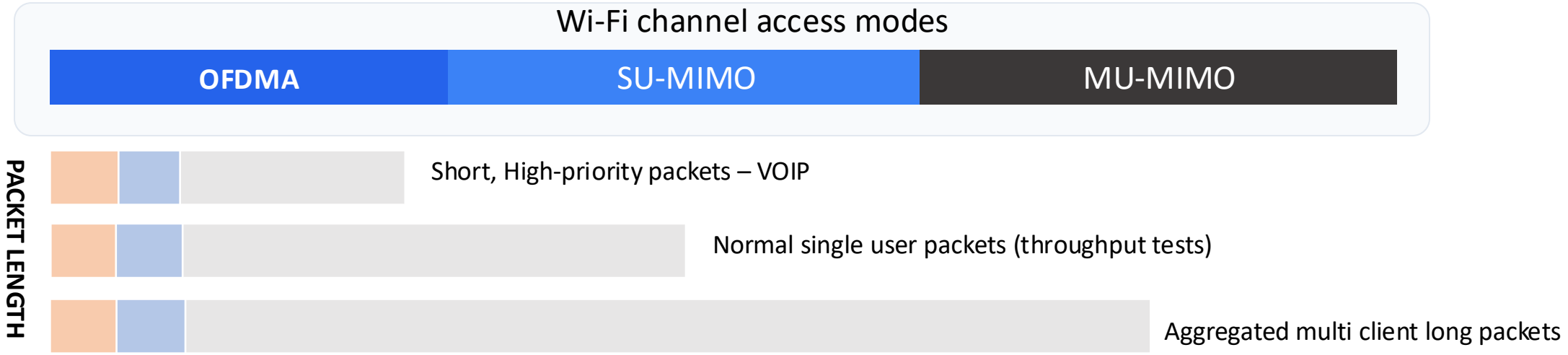


- TX BSS/RX BSS: Downlink and uplink
- OBSS: If most airtime is consumed by **OBSS**, your AP is repeatedly deferring to neighbor networks (PD detect or virtual NAV, indicating co-channel interference that requires frequency/channel reassignment.
- Interference: Measured with Energy Detect or signatures

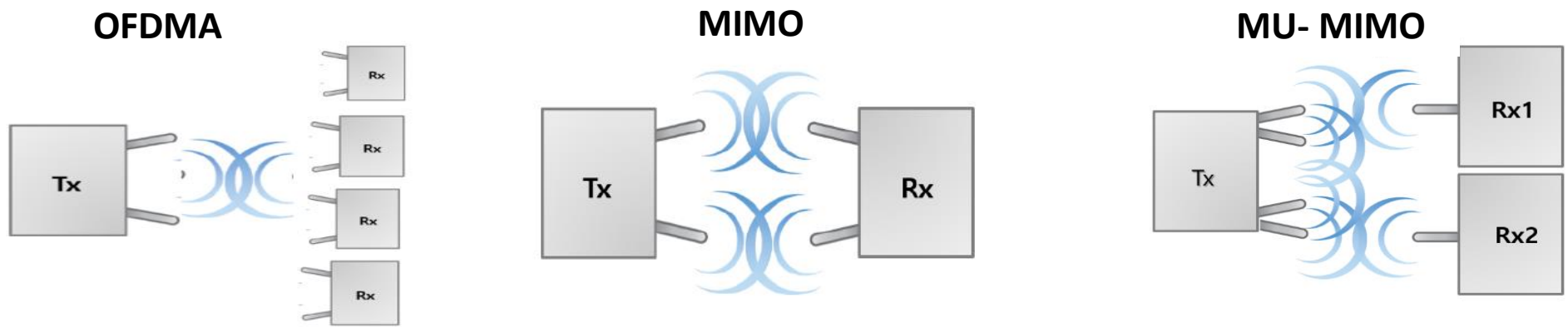
Combination of ON and OFF channel scanning measurement

Total channel busy duration directly caps maximum client capacity- Calibrate !

OFDM & OFDMA

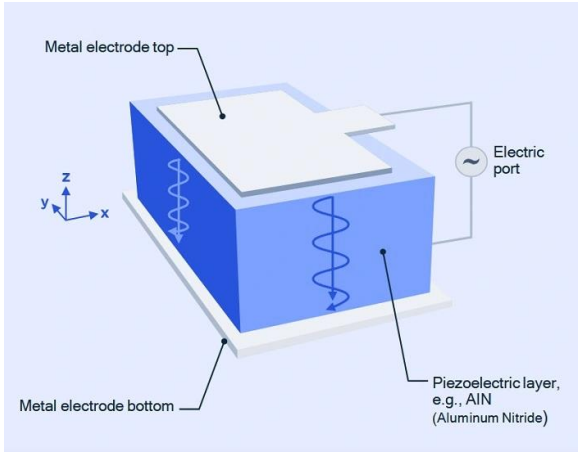


- OFDMA: multi client, simultaneous short packets transmission



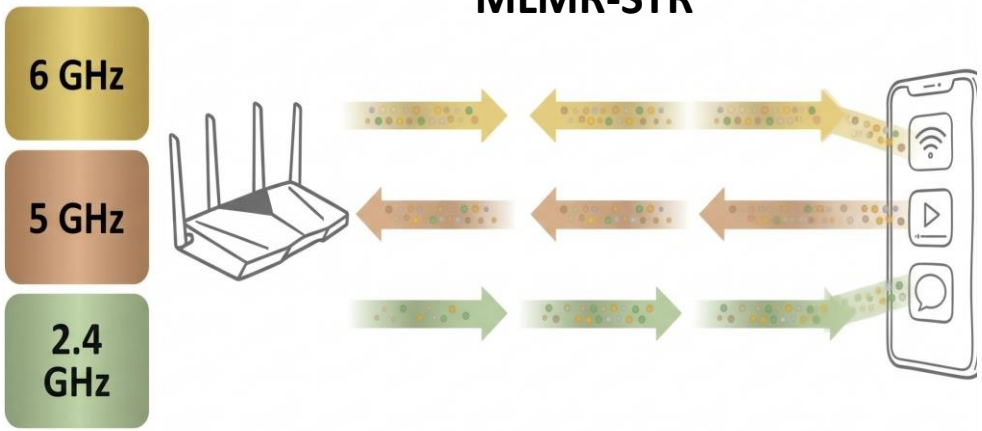
Multi Link Operation

- In-Device Coexistence (IDC) & Receiver desensitization specially for MLMR - STR mode.
- Determine if the vendor delivers true Simultaneous Transmit and Receive (STR) without cross-band desensitization, or if they silently fall back to
  - Non- Simultaneous Transmit Receive (NSTR)
  - Enhanced Multi Link Single Radio (EMLSR)

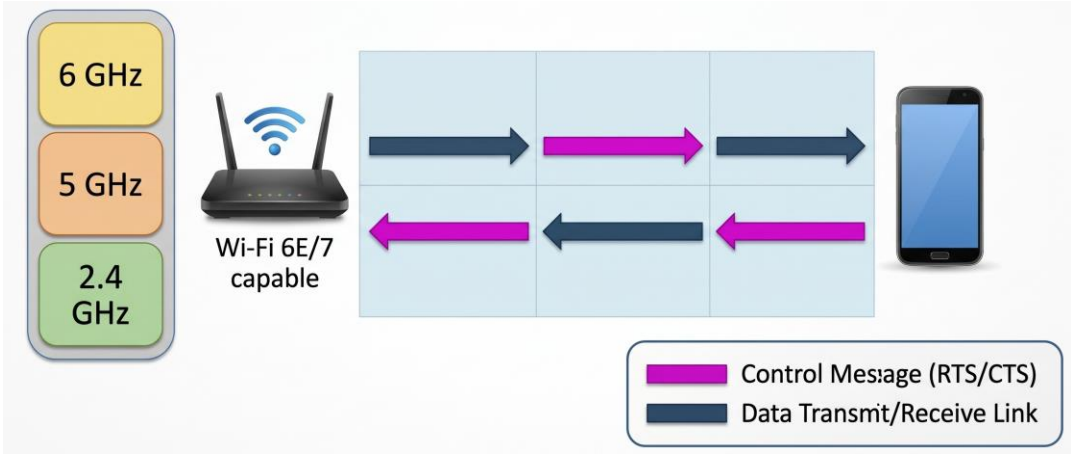


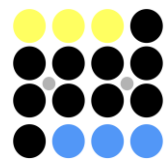
BAW filters

MLMR-STR



EMLSR





What should you ask your vendors?

https://mayursarode.github.io/Wi-Fi-HW-performance-metrics/

